Day 9 – Descriptive Statistics Primer

Data Analytics Internship | Monu Kumar Rajak | Cleaned Sample Superstore Dataset

1. Objective

Calculate and interpret mean, median, mode, standard deviation, and percentiles for key numeric columns to understand central tendency and spread.

2. Dataset and Tools

Dataset: cleaned Sample Superstore CSV. Tool: Microsoft Excel. Columns analyzed: Sales, Profit, and Quantity.

3. Summary Statistics

Statistic	Sales	Profit	Quantity
Mean	229.86	28.66	3.79
Median	54.49	8.67	3.00
Mode	Not visible	0.00	3.00
Standard Deviation	623.25	234.26	2.23
25th Percentile	17.28	1.73	2.00
50th Percentile	54.49	8.67	3.00
75th Percentile	209.94	29.36	5.00

4. Interpretation

Mean Sales (229.86) is much higher than median Sales (54.49), suggesting a right-skewed distribution where a smaller number of high-value sales raise the average. Profit mean (28.66) is also above its median (8.67), suggesting possible right skew. Quantity mean (3.79) and median (3.00) are closer together.

Standard deviation is relatively large for Sales (623.25) and Profit (234.26), showing substantial variation. The middle 50% of values range from 17.28 to 209.94 for Sales, 1.73 to 29.36 for Profit, and 2 to 5 for Quantity.

5. Conclusion

Descriptive statistics summarize typical values, variation, and distribution shape. Comparing mean with median and examining percentiles helps explain skew and spread before deeper analysis.

Note: Values are transcribed from the provided screenshot. The Sales mode was not visible; verify that cell in Excel before final submission.